Hospital Acquired (Nosocomial) Pneumonia

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College of Osteopathic Medicine

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Thank you to former scholars whose slides were modified for this lecture.

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Session Objectives

- Compare **hospital**-acquired pneumonia (HAP),and **ventilator**-acquired pneumonia (VAP).
- Explain the role of **risk factors** in different types of pneumonia
- Illustrate the **risk factors** for multidrug resistance
- Formulate a **plan** to decrease incidence of nosocomial pneumonia
- Implement and develop an algorithm for **management** of HAP based on clinical presentation

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Why is it important?

Among the **most common** hospital acquired infections (~ 22%)



Prolongs hospital stay to upto 13.1 days



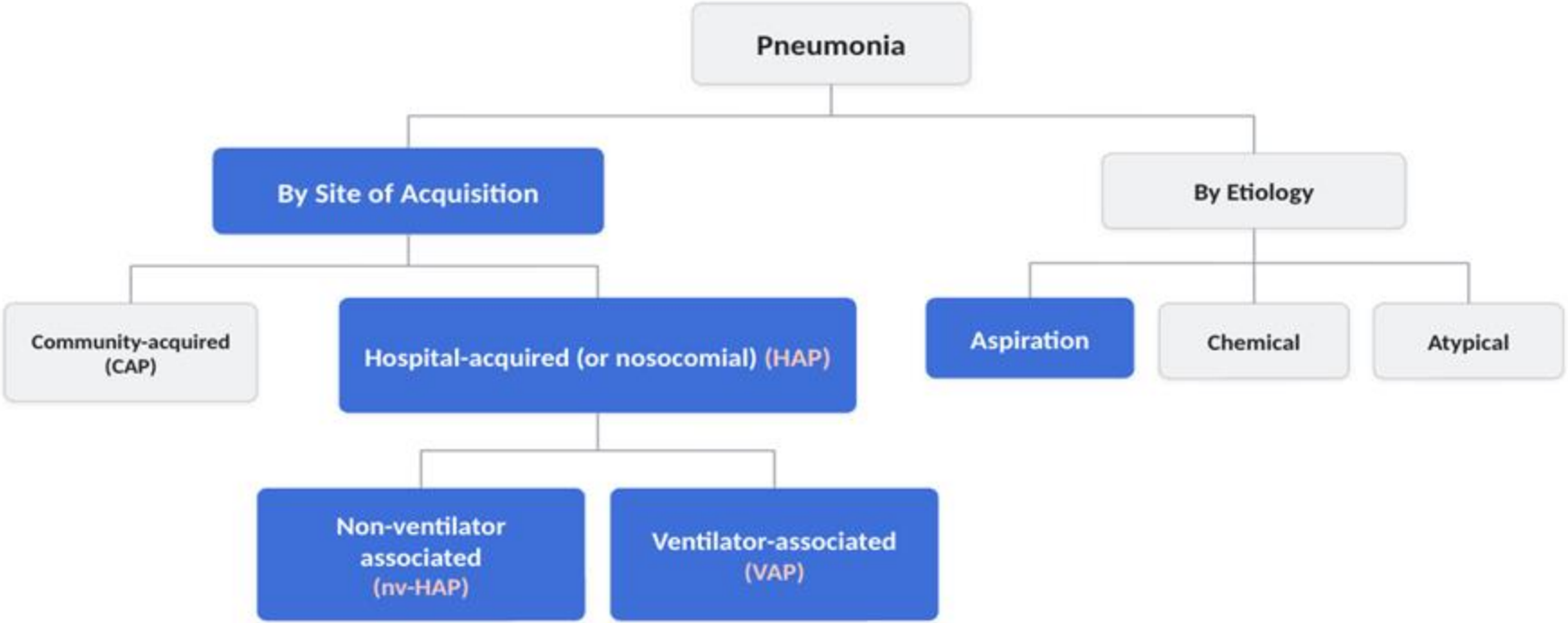
Associated with an **excess cost** of $40,000 per patient



Kalil AC, Metersky ML, Klompas M, et al. Management of adults with hospital-acquired and ventilator-associated pneumonia: 2016 clinical practice guidelines by the Infectious Diseases Society of America and the American Thoracic Society. *Clin Infect Dis.* 2016;63(5):e61-e111. doi:10.1093/cid/ciw353.https://www.idsociety.org/practice-guideline/hap_vap/.

Classification of Pneumonia

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Definitions (By Site of Acquisition)

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Type of Pneumonia	Definition
Hospital-acquired / nosocomial (HAP)	Acquired ≥ 48 hours after admission to the hospital
Ventilator-associated pneumonia (VAP)	Type of HAP acquired ≥ 48 hours after intubation on mechanical ventilator OR ≥ 48 hours following extubation
Non-ventilator-associated pneumonia (NV-HAP)	Type of HAP acquired ≥ 48 hours in hospitalized patients who are not on mechanical ventilation nor underwent extubation before pneumonia developed

OUTDATED term *Healthcare associated-pneumonia : acquired in health care facilities (eg, nursing homes, hemodialysis centers) or after recent hospitalization

Community-acquired pneumonia	Acquired outside of the hospital

Klompas, M. (2024, December 9). *Epidemiology, pathogenesis, microbiology, and diagnosis of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer. https://www.uptodate.com.nyit.idm.oclc.org/contents/epidemiology-pathogenesis-microbiology-and-diagnosis-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2&searchCorrelationId=eb1b8abe-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia

Definitions (By Etiology)

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Type of Pneumonia	Definition
Aspiration pneumonia	Acquired from entry of gastric or bacteria containing oropharyngeal fluid into the lower airway
Chemical pneumonia	Aspiration of substances (eg, acidic gastric fluid) that cause an inflammatory reaction in the lower airways, independent of bacterial infection
Atypical pneumonia	Pneumonia caused by atypical pathogens

Klompas, M. (2024, December 9). *Epidemiology, pathogenesis, microbiology, and diagnosis of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/epidemiology-pathogenesis-microbiology-and-diagnosis-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2&searchCorrelationId=b1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia

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Case 1

A 67-year-old male with a history of asthma presented to the emergency department with labored breathing and worsening respiratory distress. On physical examination, there was decreased breath sounds bilaterally. The patient was intubated and mechanically ventilated. After 72 hours, he developed a fever of 101.5°F (38.6°C). Laboratory results revealed a white blood cell count of 20,000/µL with 80% neutrophils. A chest X-ray and CT images show bilateral pulmonary infiltrate.

Based on the case details which type of pneumonia is most likely?

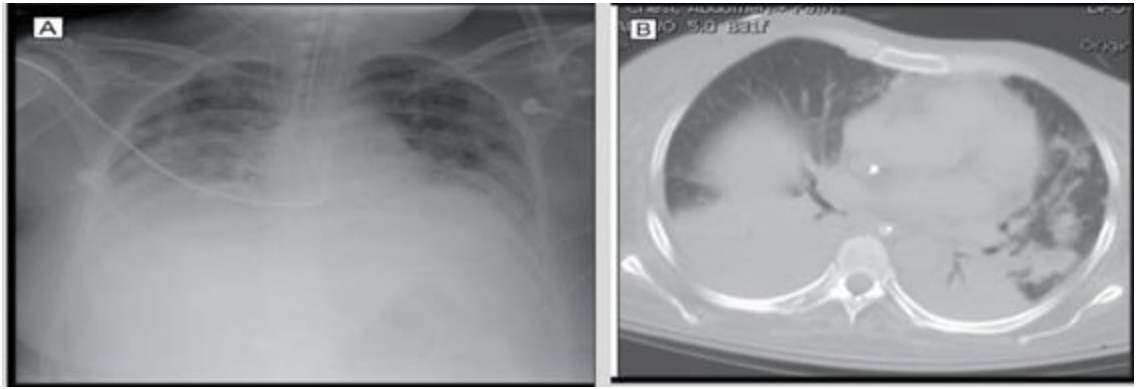
a) Community-acquired pneumonia (CAP)

b) Aspiration pneumonia

c) Health care-associated pneumonia (HCAP)

d) Viral pneumonia

e) Ventilator-associated pneumonia (VAP)



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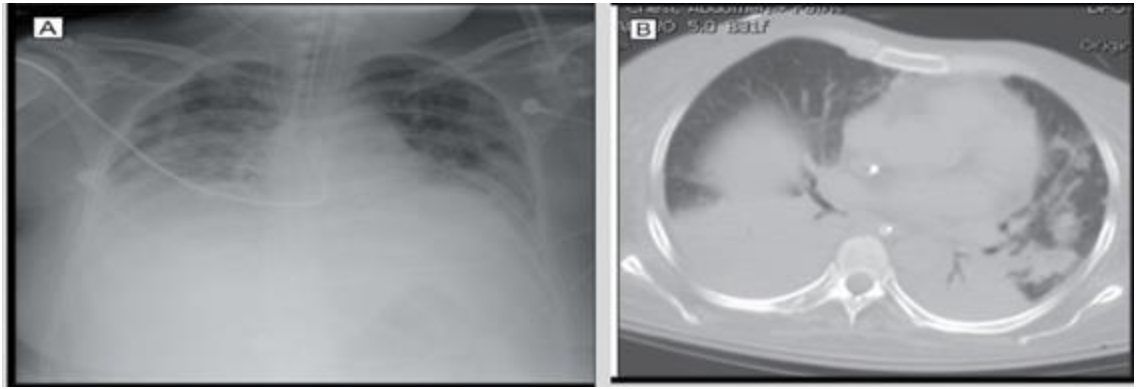
a) Community-acquired pneumonia (CAP)

b) Aspiration pneumonia

c) Health care-associated pneumonia (HCAP)

d) Viral pneumonia

e) Ventilator-associated pneumonia (VAP)



Case 1 Explanation

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a) Community-acquired pneumonia (CAP	excluded due to recent hospitalization and mechanical ventilation
b) Aspiration pneumonia	excluded due to **lack of aspiration risk factors** such as impaired swallowing, recent vomiting
c) Health care-associated pneumonia (HCAP)	excluded because it is an outdated term
d) Viral pneumonia	excluded based on neutrophilia and the clinical context ; viral does not causes significant leukocytosis and in the case it does it would show mainly lymphocytosis, not neutrophilia.
e) Ventilator-associated pneumonia (VAP)	the most likely diagnosis based on the patient's **mechanical ventilation duration, clinical signs, and radiographic evidence**

Pathogenesis and Risk Factor of HAP

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Primary route: Aspiration of microorganisms from the oropharyngeal tract into lower respiratory tract



Most significant risk factor -- **Intubation**

Endotracheal Tube (in VAP):

- Facilitates aspiration
- Formation of biofilm
- Contamination during procedure

Klompas, M. (2026, April 29). *Risk factors and prevention of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer.https://www-uptodate-com.nyit.idm.oclc.org/contents/risk-factors-and-prevention-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=3~150&usage_type=default&display_rank=3&searchCorrelationId=b1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia.

Other Risk Factors for HAP and VAP

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Impaired Host Defense/ Increased Aspiration

- Older age
- Sedation/ paralysis
- Reduced Consciousness
- Opioid use disorder
- Alcohol use disorder
- Use of glucocorticoid (immunosuppression)
- Malnutrition/chronic disease

Inoculum of organisms/ Iatrogenic

- Contaminated hospital equipment
- Poor oral hygiene
- Increased gastric pH (i.e PPI use) allows for ↑ bacterial growth

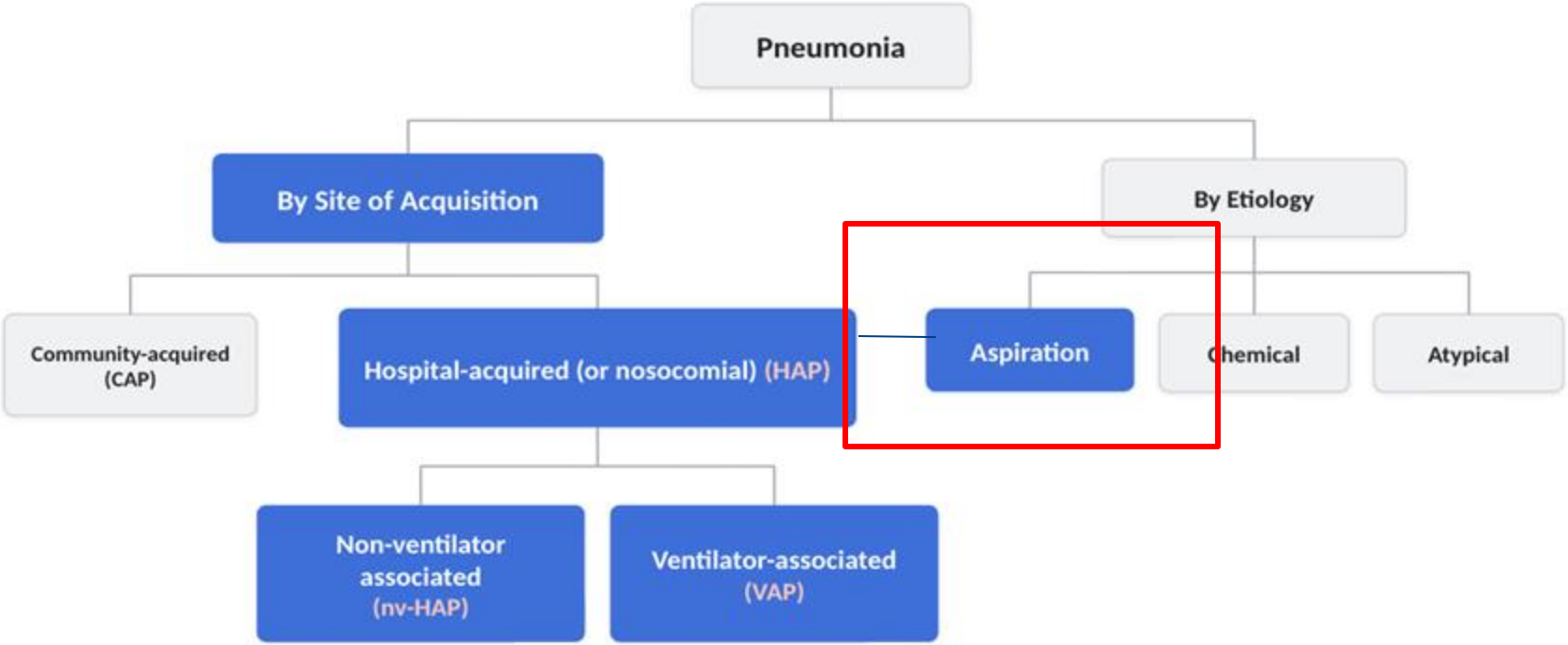
Overgrowth of virulent organisms

- Prolonged antibiotic use. **IV antibiotic use within 90 days prior to pneumonia symptoms (risk factor for drug resistant infection)**
- Number of central venous lines placed, surgeries, hospitalizations

Klompas, M. (2026, April 29). *Risk factors and prevention of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer https://www.uptodate.com.nyit.idm.oclc.org/contents/risk-factors-and-prevention-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=3~150&usage_type=default&display_rank=3&searchCorrelationId=1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTermshospital%20acquired%20pneumonia.

Classification of Pneumonia

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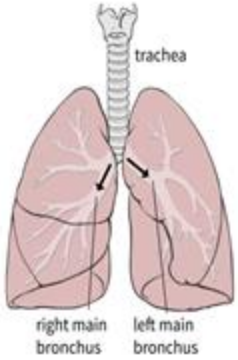


Aspiration Pneumonia as it relates to HAP

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- Aspiration of food, saliva, gastric content into the airways
- Commonly occurs after instrumentation of the upper airways or esophagus (e.g., upper GI endoscopy) or from vomiting and regurgitation of gastric contents





Right lobe steeper than Left:
Aspiration more likely on **right**

Altered consciousness
Alcoholism, seizures, head trauma, general anesthesia, drug overdose
Dysphagia
Esophageal disorders including stricture, neoplasm, diverticula, tracheoesophageal fistula, incompetent cardiac sphincter, achalasia, oropharyngeal obstruction, xerostomia
Neurologic disorder
Cerebrovascular accident, multiple sclerosis, Parkinson disease, myasthenia gravis, pseudobulbar palsy, amyotrophic lateral sclerosis
Mechanical disruption of the usual defense barriers
Nasogastric tube, endotracheal intubation, tracheostomy, upper gastrointestinal endoscopy, bronchoscopy
Other
Protracted vomiting, gastric outlet obstruction, large-volume nasogastric tube feedings, pharyngeal anesthesia, general debility, recumbent position, ascites, gastroparesis, ileus, glottic insufficiency (eg, vocal cord paralysis)

AMBOSS. (2026, June 2). *Aspiration pneumonia*. In *Pneumonia*. https://next.amboss.com/us/article/mh0Vef?q=aspiration%20pneumonia#Z9626824bc1ac2b0db5809c9fc8e9b843

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Aspiration pneumonia

Aspiration of oropharyngeal or gastric contents → pulmonary infection.

Risk factors: altered mental status (↓ cough reflex or glottic closure), dysphagia, neurologic disorders (eg, stroke), invasive tubes (eg, nasogastric tube).

Presents days after aspiration event in dependent lung segment. More common in RLL if sitting up and RUL if lying down (recumbent) due to bronchial anatomy. Can progress to abscess.

Aspiration (chemical) pneumonitis—presents hours after aspiration event. Due to gastric acid–mediated inflammation. Presents with infiltrates in lower lobe(s) and resolves with supportive treatment.

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Microbiology for HAP - The Most Common Organisms:

Organism	Classification	Things to note
S. Aureus (MSSA & MRSA)	Gram (+), coagulase (+), β-hemolytic Protein A	**Resistance** is due to altered penicillin binding proteins (PBP)
Pseudomonas aeruginosa	Gram (-) bacilli, strict aerobe, non-lactose fermenter, Oxidase & Catalase (+), motile	Frequently found in water and has a grape-like odor

Microbiology of HAP- Others include Gram -ve Bacilli

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Organism	Classification	Things to note
E. Coli	Gram (-) bacilli, Catalase (+) Lactose fermenter	K capsule antigen is an important virulence factor specifically associated with pneumonia infections
Klebsiella	Gram (-) rod & Lactose fermenter	Part of intestinal flora **ABCDE**'s (Aspiration pneumonia, a**B**scess in lungs & liver, **C**urrant jelly sputum, **D**iabetes mellitus, and **E**tOH overuse)
Enterobacter	Genus of common Gram-negative, facultatively anaerobic rods	N/A

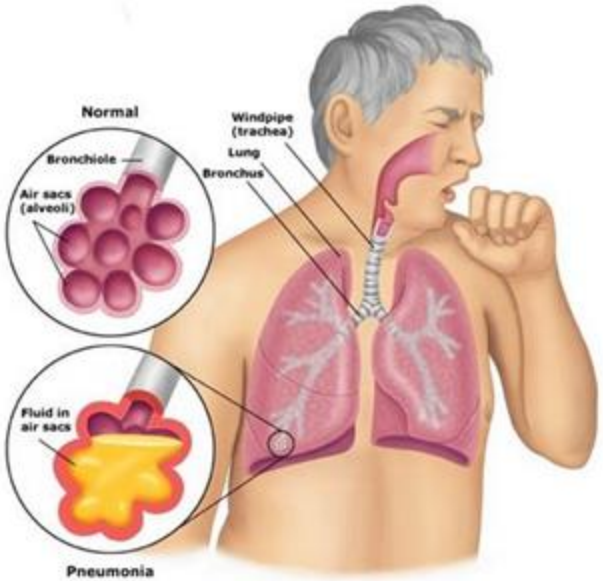
Microbiology Aspiration Pneumonia

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Organism	Classification	Things to note
Klebsiella	Gram (-) rod & Lactose fermenter	Often the cause of aspiration pneumonia in patients who have alcohol use disorder
Anaerobes	Specific species include: Peptostreptococcus, Fusobacterium, Prevotella, & Bacteroides	Slowly evolving infection eventually leads to development of pneumonia **Purple Flowers Prefer Blooming.** **Purple:Peptostreptococcus,Flowers: Fusobacterium, Prefer: Prevotella, Blooming: Bacteroides**

Clinical Presentation of HAP - Symptoms

- High grade Fever
- Productive cough
- Mucopurulent discharge
- Tachypnea and/or dyspnea
- Pleuritic chest pain



https://carlossantosmdpa.com/pnemonia.html

Kollef, M. H. (2026, April 27). *Clinical presentation and diagnostic evaluation of ventilator-associated pneumonia*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/clinical-presentation-and-diagnostic-evaluation-of-ventilator-associated-pneumonia?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=5~150&usage_type=default&display_rank=5&searchCorrelationId=b1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia.

Clinical Presentation of HAP - Signs and Physical Exam

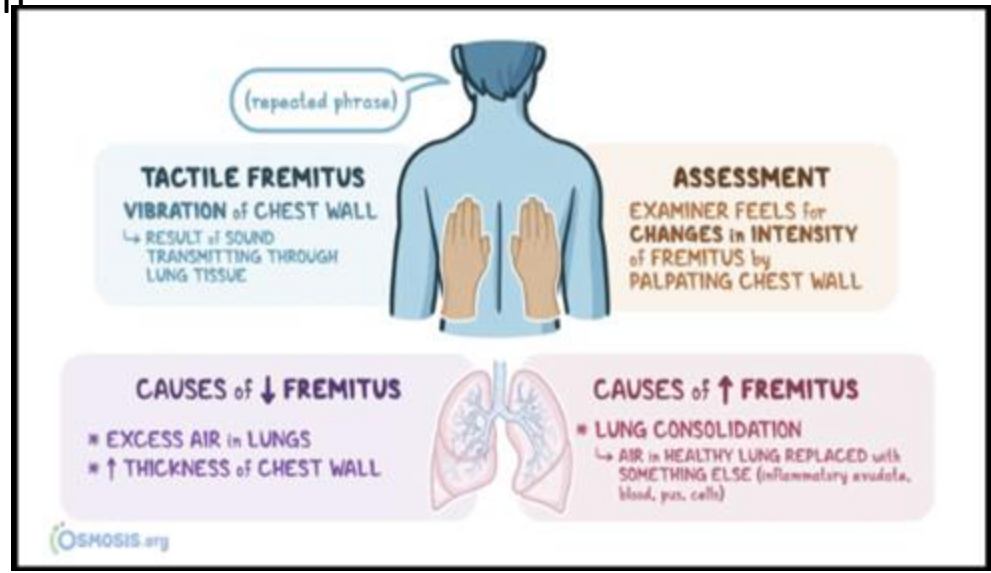
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- Hypoxemia [O2 levels below 90 indicated from pulse ox or from arterial blood draw called arterial blood gas(ABG)]
- Reduced breath sounds on auscultation
- Dullness to percussion
- Rales
- Increased tactile fremitus



Lungs Sounds

Coarse Crackles (aka Rales)



Kollef, M. H. (2026, April 27). *Clinical presentation and diagnostic evaluation of ventilator-associated pneumonia*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/clinical-presentation-and-diagnostic-evaluation-of-ventilator-associated-pneumonia?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=5~150&usage_type=default&display_rank=5&searchCorrelationId=b1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia.

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Physical findings in select lung diseases

ABNORMALITY	BREATH SOUNDS	PERCUSSION	FREMITUS	TRACHEAL DEVIATION
Pleural effusion	↓	Dull	↓	None if small Away from side of lesion if large
Atelectasis	↓	Dull	↓	Toward side of lesion
Simple pneumothorax	↓	Hyperresonant	↓	None
Tension pneumothorax	↓	Hyperresonant	↓	Away from side of lesion
Consolidation (lobar pneumonia, pulmonary edema)	Bronchial breath sounds; late inspiratory crackles, egophony, whispered pectoriloquy	Dull	↑	None

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Diagnosis of HAP

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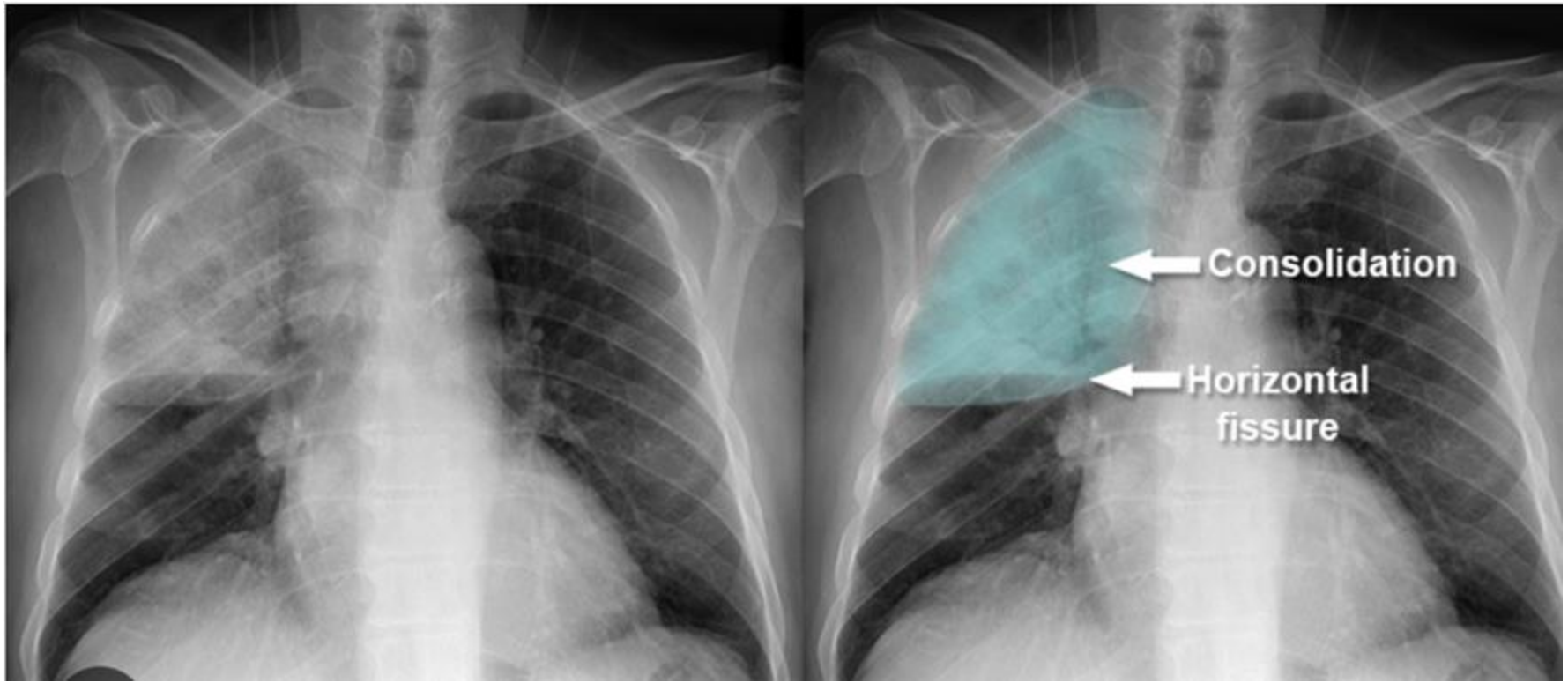
- Blood culture (prior to antibiotics)
- Sputum culture (prior to antibiotics)
- Bloodwork (CBC, CMP, procalcitonin) : leukocytosis on CBC
- Lower respiratory tract sample especially for VAP (prior to antibiotics):
  - bronchoalveolar lavage (invasive)
  - endotracheal/tracheobronchial aspirate sampling (non invasive)
- Chest X ray for HAP: **solid consolidation** (known as lobar pneumonia) or **patchy infiltrates** (known as bronchopneumonia) or **Interstitial infiltrates** (known as interstitial pneumonia)

Kollef, M. H. (2026, April 27). *Clinical presentation and diagnostic evaluation of ventilator-associated pneumonia*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/clinical-presentation-and-diagnostic-evaluation-of-ventilator-associated-pneumonia?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=5~150&usage_type=default&display_rank=5&searchCorrelationId=b1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia.

First Aid (2026, June). Pneumonia. https://usmle-rx.scholarrx.com/high-yield-topics?id=1381.

HAP Chest X ray: Lobar Pneumonia Solid Consolidation

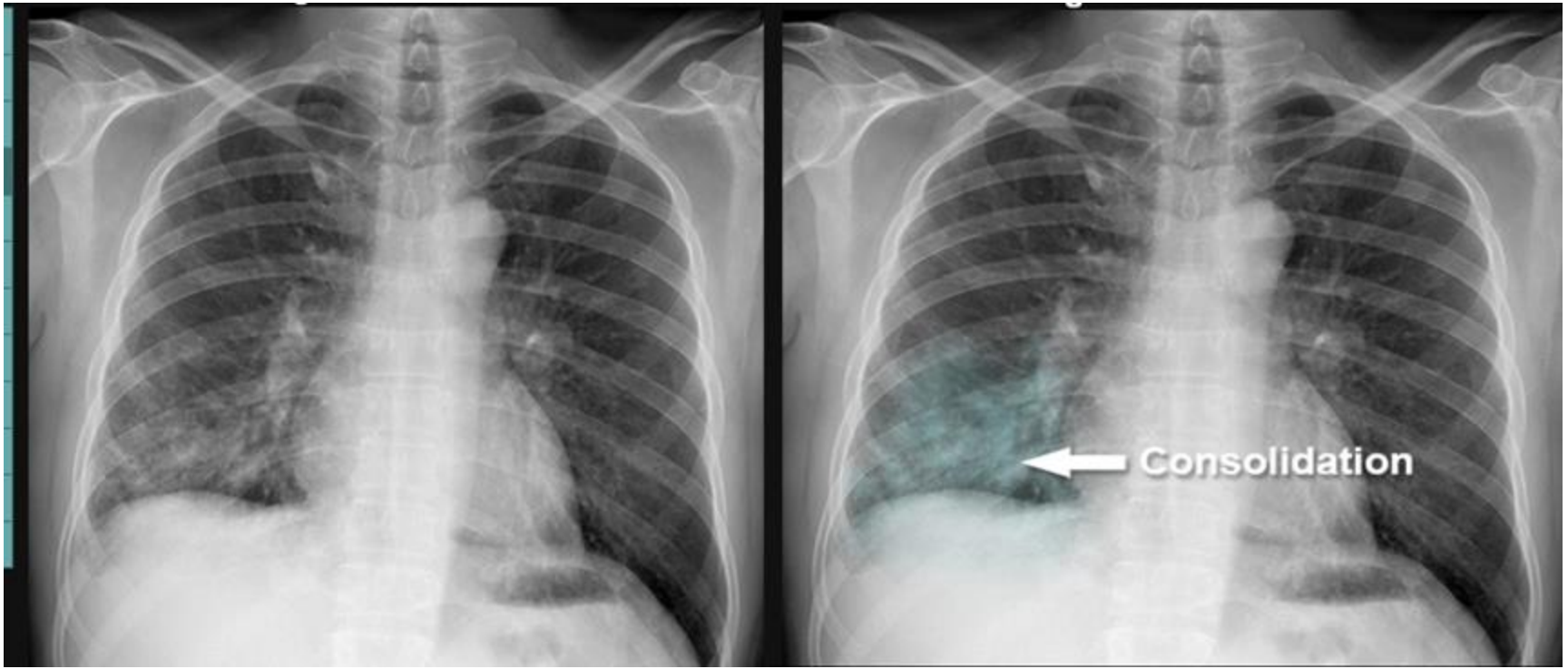
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https://www.radiologymasterclass.co.uk/gallery/chest/pulmonary-disease/consolidation_lobar.

HAP Chest X ray: Bronchopneumonia Patchy Infiltrates

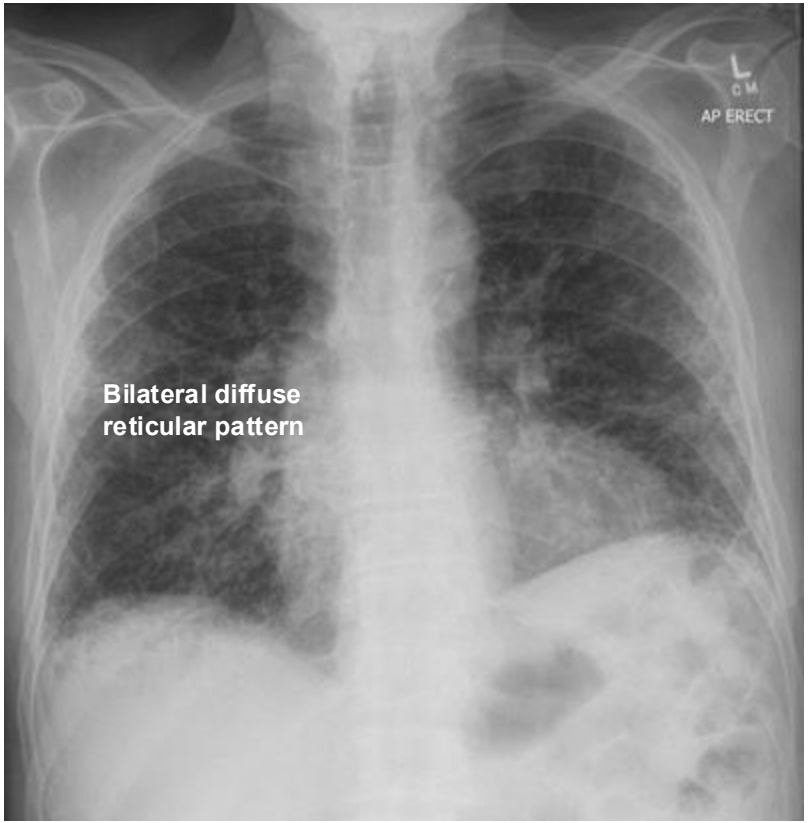
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https://www.radiologymasterclass.co.uk/gallery/chest/pulmonary-disease/lung_zones.

HAP Chest X ray: Interstitial “Ground Glass” Infiltrates

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https://radiopaedia.org/articles/usual-interstitial-pneumonia.

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Pneumonia

TYPE	TYPICAL ORGANISMS	CHARACTERISTICS
Lobar pneumonia 	*S pneumoniae* (most common), *Legionella*, *Klebsiella*.	Intra-alveolar exudate → consolidation A; may involve entire lobe or the whole lung.
Bronchopneumonia	*S pneumoniae*, *S aureus*, *H influenzae*, *Klebsiella*.	Acute inflammatory infiltrates from bronchioles into adjacent alveoli; patchy distribution involving ≥ 1 lobe.
Interstitial (atypical) pneumonia 	*Mycoplasma*, *Chlamydophila pneumoniae*, *Chlamydophila psittaci*, *Legionella*, *Coxiella burnetii*, viruses (RSV, CMV, influenza, adenovirus).	Diffuse patchy inflammation localized to interstitial areas at alveolar walls; CXR shows bilateral multifocal opacities B. Generally follows a more indolent course ("walking" pneumonia).

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Treatment

Antibiotics! (Pharmacology discussed in a later slide)

Things to Consider

- Risk of multi-drug resistance (MDR) pathogens
- Local antibiotics resistance
- Individual patient's prior microbiology data



Drug Resistance

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Multi-drug resistant (MDR): resistance of organisms to at least one agent in three different antibiotic drug classes

Extensive drug resistant: resistance to at least one agent in all but two antibiotic classes



Pan-drug resistance: resistant to all agents in all antibiotic classes

Klompas, M. (2024, December 9). *Epidemiology, pathogenesis, microbiology, and diagnosis of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/epidemiology-pathogenesis-microbiology-and-diagnosis-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2&searchCorrelationId=b1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia

Identifying Drug Resistance in Hospitals

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Risk factors for MDR gram-negative bacteria (pseudomonas and others):

Patient had prior colonization with MDR gram-negative bacteria on culture

They are receiving treatment in a hospital unit (or region) in which >10 percent of gram-negative bacterial isolates are MDR

Risk factors for methicillin-resistant *Staphylococcus aureus* (MRSA):

•Patient had prior colonization with MRSA on culture

•Treatment in a hospital unit (or region) in which >20 percent of *S. aureus* isolates are resistant to methicillin

Bottomline: Awareness of local resistance patterns is critical for decisions regarding empiric therapy for HAP and VAP (Using Hospital Antibiograms)

Klompas, M. (2024, December 9). *Epidemiology, pathogenesis, microbiology, and diagnosis of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/epidemiology-pathogenesis-microbiology-and-diagnosis-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2&searchCorrelationId=b1b8abae-39f1-464e-b29b-

Hospital Antibiogram

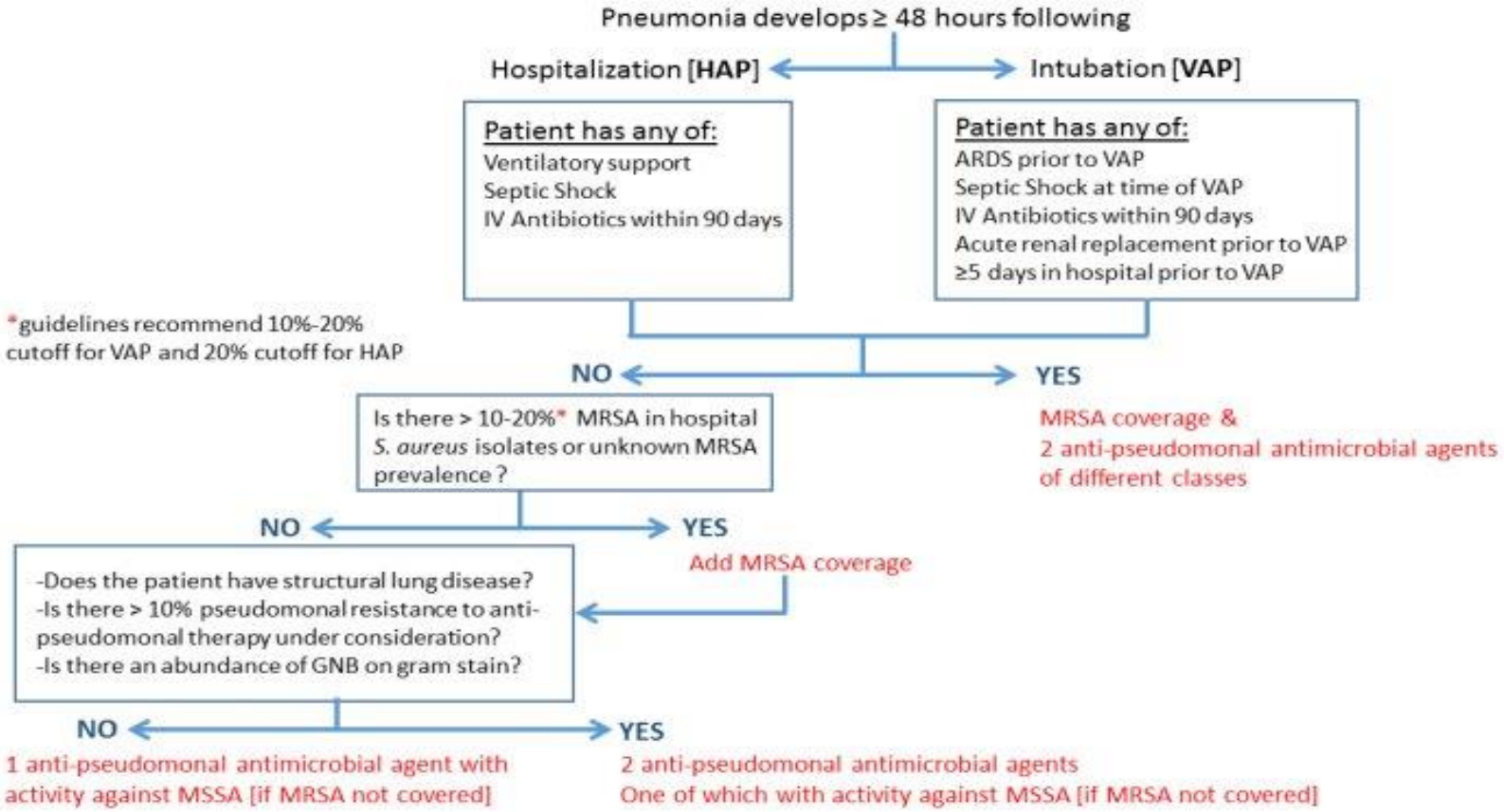
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Resistance of common bacterial isolates, St Teresa's Hospital, 2018																						
	% nonsusceptible																					
Organism	No. of isolates tested	Amikacin	Ampicillin	Amoxicillin+clavulanate	Cefotaxime	Ceftazidime	Ceftriaxone	Cefuroxime (IV)	Cephalothin	Chloramphenicol	Ciprofloxacin	Clindamycin	Cotrimoxazole	Erythromycin	Gentamicin	Imipenem	Levofloxacin	Linezolid	Methicillin #	Penicillin	Piperacillin+tazobactam	Nitrofurantoin *	Tetracycline	Vancomycin
Acinetobacter species	45	0				20					16				11	13					16			
Escherichia coli	963-1327		72	15				27	32		30		42		27							2		
Enterococcus species	99-162		9														27						80	4
Haemophilus influenzae	432		39	6				6					47											
Klebsiella species	494		100	14				19	21				19		9									
Pseudomonas aeruginosa	244	2				6					12				5	12					7			
Salmonella species	238-356		58			5	5				65		24											
Staphylococcus aureus	855-867												1	29	4				18	81				0
MSSA	703-714												0	24	3				0	77				0
MRSA	152-153												3	53	12				100	100				0
Streptococcus agalactiae	181-221											54		54			20			0				0
Streptococcus pneumoniae &	159-160			6	3		4	26					49	81						1			75	0

https://impact.chp.gov.hk/antibiograms_StT.php.

Treatment Algorithm

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Treatment (Basic Idea)

Blood /Sputum Culture

↓

Start Empiric therapy **right away** (taking into account hospital antibiograms and local resistance patterns)

↓

De-escalate empiric therapy **ASAP** when causative agent found (confirmed by blood/sputum culture) or when clinical improvement seen

Klompas, M. (2026, March 17). *Treatment of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/treatment-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1&searchCorrelationId=b1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerm=hospital%20acquired%20pneumonia.

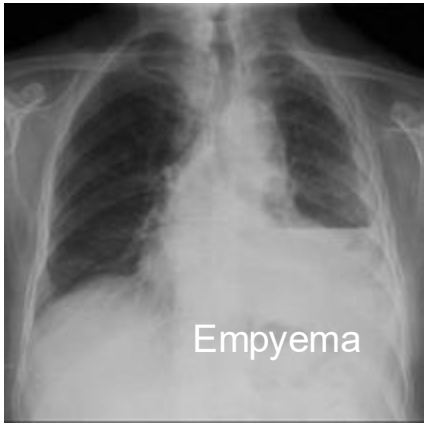
Treatment (Pharmacology)

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Pneumonia Type	Pathogen	Antibiotic
HAP or VAP	MRSA	Vancomycin or Linezolid
	Pseudomonas	Piperacillin/Tazobactam or Cefepime (or Imipenem)
Aspiration	Anaerobes	**Clindamycin**, Amoxicillin/clavulanate, or Metronidazole & a Beta Lactam

Complications for HAP

- Pleural effusion - fluid in pleural space due to increased capillary permeability
- Empyema - pus in the pleural space from bacterial invasion
- Lung Abscess - infected cavity inside the lung parenchyma/tissue
- Sepsis- high fever, hypotension, tachycardia , unstable
- Respiratory Failure
- Renal Failure





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Prevention of HAP

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Prevent Aspiration

- Elevate the head of bed to 30 - 45°
- Minimize sedation/ sedation interruption
- Frequent subglottic drainage
- Ensuring proper swallow using speech and swallow evaluation, nausea control etc

Decontamination of Oropharynx/Digestive tract

Maintain Oral Care and Toothbrushing

Enteral nutrition over parenteral nutrition if oral intake not possible

VAP specific Bundle

Avoid intubation/ re-intubation when possible
Facilitate early extubation
Implement proper weaning methods: spontaneous breathing trials
Semi Recumbent positioning
Maintain physical conditioning: mobilizing programs

Klompas, M. (2026, April 29). *Risk factors and prevention of hospital-acquired and ventilator-associated pneumonia in adults*. UpToDate. Wolters Kluwer. https://www-uptodate-com.nyit.idm.oclc.org/contents/risk-factors-and-prevention-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults?search=hospital%20acquired%20pneumonia&source=search_result&selectedTitle=3~150&usage_type=default&display_rank=3&searchCorrelationKeb1b8abae-39f1-464e-b29b-a3de8e3b664f&searchCorrelationTerms=hospital%20aquired%20pneumonia.

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Cases

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Case 2

A 55-year-old female with past medical history of COPD ,recent hospitalization for UTI that was treated with antibiotics and 50-pack year history (quit 3 years ago), presents to the emergency department with progressive dyspnea. Patient was admitted to the medical floor for acute COPD exacerbation and treated with steroids and albuterol nebulizer. Fifty-six hours later, patient started to develop fever and decreased oxygen saturation.The following CXR is obtained.

What are the most important predictors of nosocomial pneumonia in this patient?



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What are the most important predictors of nosocomial pneumonia in this patient?

- Patient developed signs and symptoms **> 56 hrs after hospitalization**
- Recent use of antibiotics (for UTI)
- Comorbidity (COPD)



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Case 3

A sputum sample from a patient with suspected pneumonia is sent to the lab. Gram stain is shown. The bacteria is also catalase (+) and coagulase (+). Which of the following microbes is the most likely causative agent?

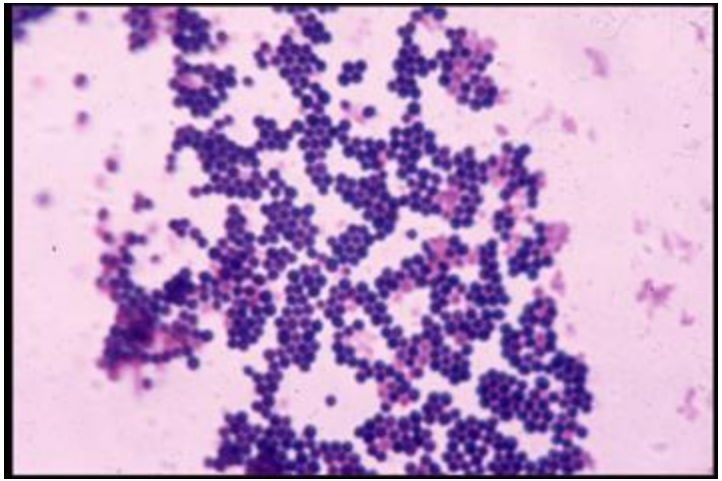
a) S. aureus

b) S. pneumoniae

c) Klebsiella

d) S. epidermidis

e) S. pyogenes



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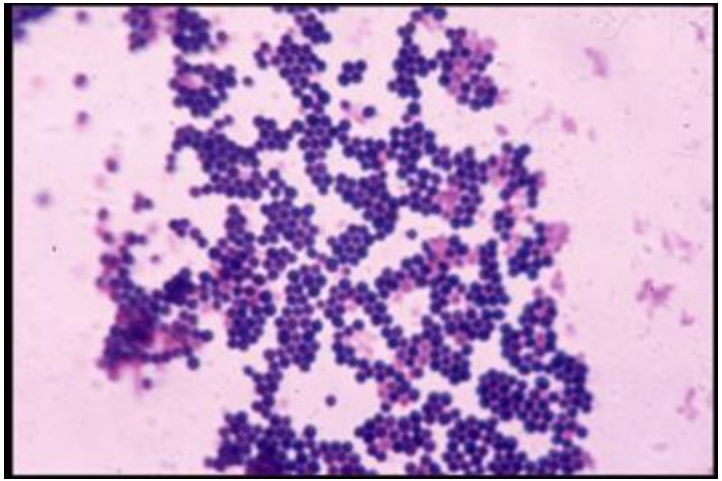
a) S. aureus

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Case 3 Explanation

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a) S. aureus	**the most likely causative agent because it is a gram-positive cocci in clusters, catalase-positive, and coagulase-positive**
b) Klebsiella	excluded due to its gram-negative rod morphology
c) S. pneumoniae	excluded because it is catalase-negative and typically appears as diplococci
d) S. pyogenes	excluded because it is catalase-negative and appears in chains
e) S. epidermidis	excluded because it is coagulase-negative

Case 4

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Mr. Smith is a 65-year-old male with a PMH of HTN, HLD, and asthma presents to the ED by ambulance with a 2-week history of progressive SOB and productive cough with yellow sputum. Pt has a 40-pack year smoking history and alcohol substance use disorder. His vitals are BP 100/56 mmHg, HR 95 bpm, RR 21 bpm and Temp 101.1°F. He is admitted to the ICU for a presumed sepsis secondary to pneumonia. He is treated with albuterol, IV fluids, and antibiotics.

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Case 4

You obtain a CXR for Mr. Smith shown on the right.

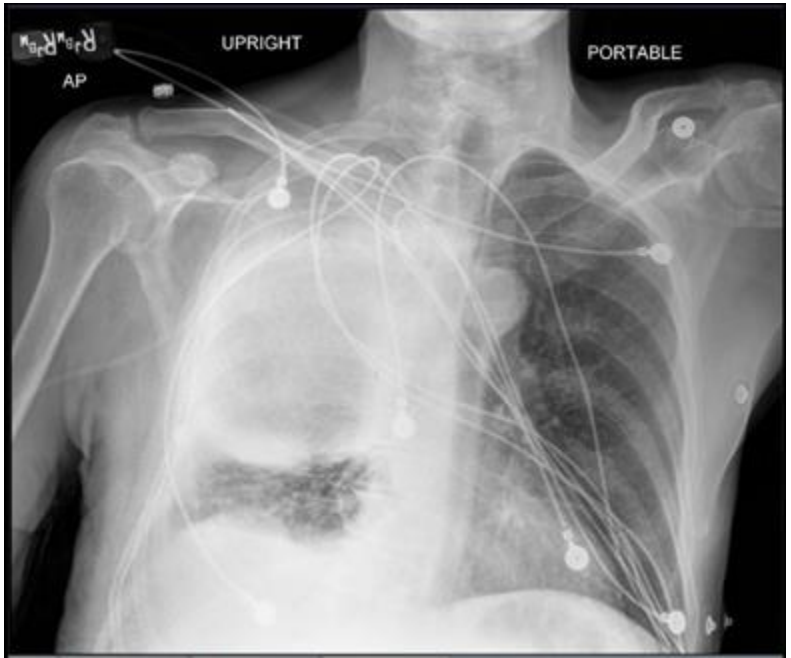
What is the most likely classification of Mr. Smith's pneumonia?

a) Community-acquired pneumonia

b) Hospital-acquired pneumonia

c) Aspiration pneumonia

d) Ventilator-associated pneumonia



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Case 4

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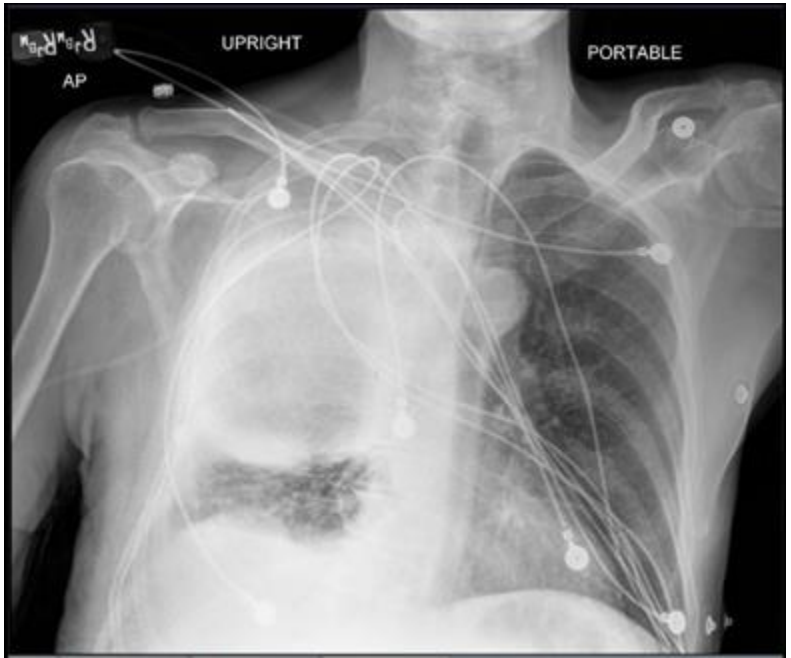
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Case 4

You obtain a CXR for Mr. Smith shown on the right.

What is the most likely pathogen?

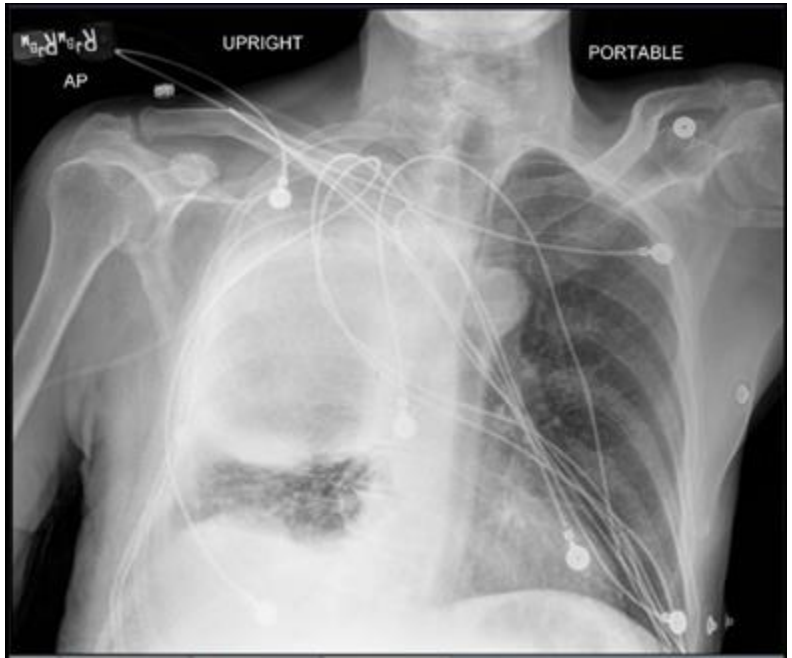
a) Mycobacterium tuberculosis

b) C. difficile

c) N. meningitidis

d) S. pneumoniae

e) Klebsiella



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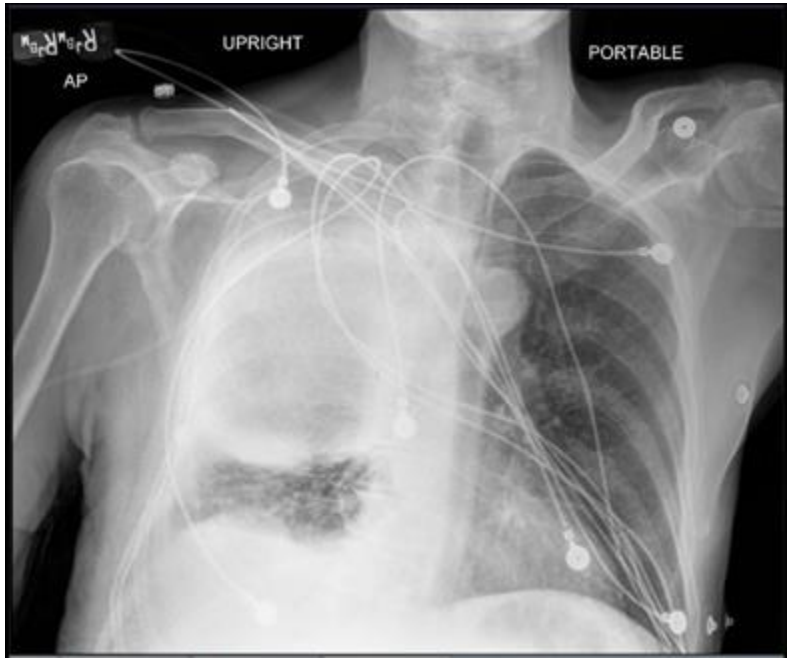
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b) C. difficile

c) N. meningitidis

d) S. pneumoniae

e) Klebsiella



Case 4 Explanation

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What is the most likely classification of Mr. Smith's pneumonia?

a) Community-acquired pneumonia	Excluded because his history of alcohol use disorder and a significant smoking history increase the risk for aspiration pneumonia
b) Hospital-acquired pneumonia	Excluded because Mr. Smith did not acquire the pneumonia while in a healthcare setting; he presented from the community
c) Aspiration pneumonia	**Correct answer due to his alcohol use disorder and significant smoking history, which increase the risk of aspiration**
d) Ventilator-associated pneumonia	Excluded because Mr. Smith is not on mechanical ventilation.

Case 4 Explanation

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What is the most likely pathogen?

a) Mycobacterium tuberculosis	Excluded because TB typically presents with a more chronic course and is less commonly associated with aspiration pneumonia
b) Clostridium difficile	Excluded because it is primarily associated with antibiotic-associated diarrhea and colitis, not pneumonia
c) Neisseria meningitidis	Excluded because it primarily causes meningitis and septicemia rather than pneumonia
d) Streptococcus pneumoniae	Excluded as it is more commonly associated with community- acquired pneumonia rather than aspiration pneumonia in patients with his risk factors
e) Klebsiella pneumoniae	**Correct answer as it is commonly associated with aspiration pneumonia, especially in patients with chronic alcoholism**

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Case 5

Mrs. Puff is a 75-year-old female with a PMH of COPD, DM type 2 and obesity status post cholecystectomy 5 days ago. Overnight was noted to have fever, chills and productive cough with yellow phlegm. Vitals show HR 120 bpm, RR 30 bpm, BP 80/50 mmHg, Temp 101.3°F, O2 saturation 84% on room air. On physical exam pt shows cyanosed lips, labored breathing, and use of accessory muscles of respiration. Lung auscultation shows decreased breath sounds on the lower right lung, increased vocal resonance and coarse rales on the rig



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Case 5

Chest x-ray is consistent with right lower lobe infiltrate. Which of the following would be the most

appropriate empiric antibiotic for this patient?

a) Piperacillin-tazobactam

b) Imipenem, aminoglycoside and vancomycin

c) Ceftriaxone and macrolide

d) Metronidazole

e) Amoxicillin



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Case 5 Explanation

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a) Piperacillin-tazobactam	Excluded: While it covers Pseudomonas, it might not cover MRSA, which is a concern given her recent surgery and severe presentation
b) Imipenem, aminoglycoside and vancomycin	**Correct: This combination covers resistant Gram-negative bacteria and MRSA, appropriate for severe hospital acquired pneumonia in a patient with multiple comorbidities and recent hospitalization**
c) Ceftriaxone and macrolide	Excluded: Suitable for community-acquired pneumonia but lacks adequate coverage for resistant hospital-acquired pathogens likely in this patient
d) Metronidazole	Excluded: Targets anaerobes, not suitable for the bacterial pneumonia suggested by the clinical presentation and imaging
e) Amoxicillin	Excluded: Effective for mild community-acquired pneumonia, but not adequate for severe hospital acquired pneumonia in a patient with significant risk factors

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Summary

- HAP → Pneumonia diagnosed ≥48 hours after hospital admission
- VAP → Type of HAP diagnosed ≥48 hours after endotracheal intubation
- Greatest risk factor for developing VAP is intubation
- Greatest risk of MDR pneumonia is the use of IV antibiotics within 90 days prior to diagnosis
- Common bugs that cause HAP & VAP → S. Aureus & Pseudomonas
- Common bugs that cause aspiration pneumonia →Anaerobes & Klebsiella
- PE findings associated with pneumonia → dullness to percussion & ↑ tactile fremitus
  - Radiographic patterns of pneumonia →lobar, bronchopneumonia, & interstitial
- No risk for MDR organisms → empiric therapy covering S. aureus, P. aeruginosa &other gram (-) bacilli

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Thank You!

Questions?

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